

1. What are the three main layers of software architecture? Write one line about each.

The standard 3-tier architecture consists of the following functional layers:

- **Presentation Tier:** This is the topmost level of an application, serving as the user interface that translates tasks and results into something the user can understand.
- **Logic Tier:** This middle layer controls the application's functionality by performing detailed processing and managing the transfer of data between the surrounding layers.
- **Data Tier:** This foundational level manages data persistence and includes the actual database servers or file systems where information is stored and retrieved.

2. Define client and server in your own words with one real-life example.

In a networked environment, these roles define how computers interact with one another:

- **Client:** A client is a program or device that initiates requests for services or resources from another computer.
- **Server:** A server is a specialized device or program that listens for those requests and provides the necessary functionality or data to the client.
- **Real-Life Example:** When you use a web browser (Client) on your laptop to visit a website, the browser sends a request to the website's host computer (Server), which then sends back the files required to display the page.

3. Write 2 differences each between desktop apps, web apps, and mobile apps.

Software applications vary significantly based on their deployment platform and accessibility:

Desktop Apps	Installed directly on a specific computer's	Often designed to work offline and utilize
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	operating system (e.g., Windows, macOS).	high-performance local hardware resources.
Web Apps	Accessed via a web browser and do not require local installation.	Generally require a continuous internet connection and are compatible across different platforms.
Mobile Apps	Specifically developed for portable devices like smartphones and tablets.	Can utilize unique device features such as GPS, cameras, and push notifications.

4. Name any 3 databases and mention whether each is SQL or NoSQL.

Modern software relies on various database management systems categorized by how they structure data:

1. **MySQL:** A widely used **SQL** (Relational) database.
2. **MongoDB:** A popular **NoSQL** (Non-relational) database that stores data in flexible, JSON-like documents.
3. **PostgreSQL:** An advanced, open-source **SQL** (Relational) database known for its robustness.